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Valuing American Equity Options with a Stochastic Interest Rate: A Note

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ABSTRACT

This note presents a simple, efficient numerical procedure to price American-style equity options with stochastic interest rates. The procedure involves transforming the original stock and interest rate processes to obtain two new independent processes. A three-dimensional lattice can then be built to price any options written on the stock. The lattice is simple and efficient, because each node on the lattice branches into only four nodes, despite the presence of two stochastic variables.

Practitioners can use the procedure to price long-term options such as LEAPS, and researchers can employ the procedure to conduct empirical studies where a stochastic interest rate must be assumed.

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I. INTRODUCTION

Recently, Long-term Equity Anticipation Securities (LEAPS) have been formally listed on major exchanges such as CBOE, the American Stock Exchange, the Pacific Stock Exchange, and the Philadelphia Stock Exchange. Currently, LEAPS are available on more than 100 individual stocks and on the S&P 100 and S&P 500 indexes. LEAPS are long-term American-style equity options whose time-to-maturity is at least 2 years at the time of issue. This longer maturity makes it inappropriate to assume a constant interest rate as in the case of conventional short-term exchange-traded options. A stochastic interest rate is necessary to capture the reality. Although a closed-form formula exists for a European option with a stochastic interest rate (Merton, 1973), its American counterpart does not exist. We have to resort to a multi-variable numerical procedure. Several authors have proposed such procedures [e.g., Boyle (1988) and Boyle, Evnine, and Gibbs (1989)] to price American-style options. Unfortunately, these procedures all require a constant interest rate. As a result they are not applicable to American-style equity options with a stochastic interest rate. There are two exceptions. Hull and White (1990) proposed a finite difference approach which is essentially a trinomial lattice. Their approach can be used to price American equity options with a stochastic interest rate. To implement the model, we need to build a tree on which each node will emanate into nine nodes, because there are two state variables involved. Kishimoto (1989) developed a general numerical framework by combining the Ho-Lee (Ho and Lee, 1986) term structure model with the Cox-Ross-Rubinstein (Cox, Ross, and Rubinstein, 1979) binomial lattice. In this model, although each node emanates into only four nodes, the programming is extremely complex since it involves the method of alternating directions and the fitting of the term structure. The purpose of this note is to show a very simple, efficient procedure for pricing American equity options with a stochastic interest rate. The procedure is based on Nelson and Ramaswamy (1990). The remainder of the paper is organized in two sections. Section II describes the procedure and Section III concludes the note.

II. A THREE-DIMENSIONAL BINOMIAL PROCEDURE

Nelson and Ramaswamy (1990) show that almost all diffusions can be approximated with a simple binomial process. The key is to transform the original process into a constant variance process, which in turn can be approximated with a simple binomial process. An option based on the original process can then be priced with the binomial tree. Nelson and Ramaswamy identify as examples the

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binomial processes for lognormal and constant elasticity variance (CEV) processes for a stock, and the Ornstein-Uhlenbeck and CIR (Cox, Ingersoll, and Ross, 1985a) processes for an interest rate. For brevity we will not restate the converting procedures here. Instead, we will illustrate how the Nelson-Ramaswamy technique can be generalized to price American-style equity options with a stochastic interest rate. We will first outline the four major steps and then illustrate the procedure with an example.

Once we have assumed the stock price process (e.g., CEV) and the interest rate process (e.g., CIR), we could price an American option on the stock in four steps.

- 1) Transform the stock price and interest rate processes into constant-variance processes.
- 2) Define a new variable as a function of the transformed variables so that the new variable is orthogonal to the constant-variance interest rate.
- 3) Model the constant-variance interest rate process and the newly defined orthogonal process as two binomial processes, respectively; the two trees are combined via four joint probabilities.
- 4) At each node on the tree, convert the variables back to the original interest rate and the stock price; the American option is then priced by backward deduction.

To better understand this technique, we will examine an example. Suppose the stock price (denoted by S) follows a geometric Brownian motion and the interest rate (denoted by r) follows an Ornstein-Uhlenbeck process. Then the risk-neutral process for the stock price is:

$$\frac{dS}{S} = r(t)dt + \sigma_s dz(t) \quad (1)$$

where σ_s is the volatility of the stock price and $z(t)$ is a standard Wiener process. The risk-neutral process for the interest rate is:¹

$$dr(t) = \{k[\mu - r(t)] + \lambda\sigma_r\} + \sigma_r dw(t) \quad (2)$$

where k is the reversion speed, μ is the long-term reversion target, σ_r is the interest rate standard deviation, λ is the market price of interest rate risk, and $w(t)$ is a standard Wiener process whose instantaneous correlation with $z(t)$ is ρ . We assume that k , μ , σ_s , σ_r , λ , and ρ are constants.

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The first step is to transform the processes in equations 1 and 2 into constant variance processes. To simplify programming, it is preferable to have unit variance processes. To this end, we let $X = \ln S / \sigma_s$ and $R(t) = r(t) / \sigma_r$. By Ito's lemma:

$$dX = \frac{R(t)\sigma_r - \sigma_s^2/2}{\sigma_s} dt + dz(t) \quad (3)$$

$$dR(t) = \{k[\mu/\sigma_r - R(t)] + \lambda\} dt + dw(t) \quad (4)$$

The second step is to define a new process which is orthogonal to $R(t)$. It can be shown that the following variable meets the requirements:²

$$Y(t) = \frac{X - \rho R(t)}{\sqrt{1 - \rho^2}} \quad (5)$$

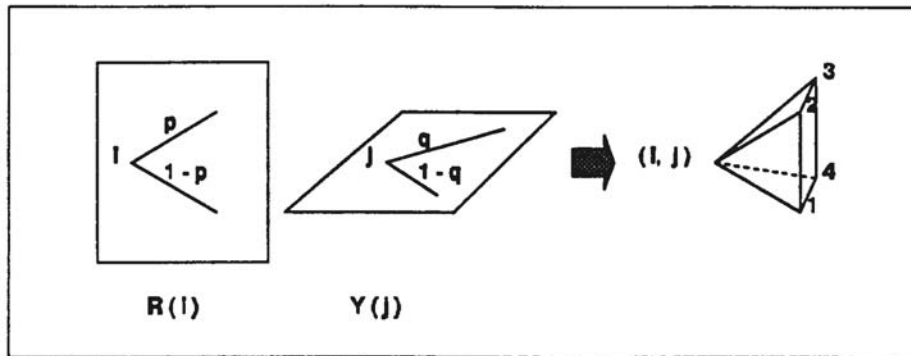
By Ito's lemma, the process for $Y(t)$ can be written as:

$$dY(t) = \frac{\left(\frac{\sigma_r}{\sigma_s} + \rho k\right)R(t) - \frac{\rho}{\sigma_r}(k\mu + \lambda\sigma_r) - \frac{\sigma_s}{2}}{\sqrt{1 - \rho^2}} dt + dv(t) \quad (6)$$

where $v(t)$ is a standard Wiener process orthogonal to $R(t)$.

The third step is crucial. We will build a three-dimensional lattice with each node branching into four. Specifically, we model equations 4 and 6 as two independent binomial processes and then calculate the joint probability for each branch. Referring to diagram 1, we can see that at a given node (i, j) , $R(t_i)$ and $Y(t_j)$ each has two branches. Therefore, from node (i, j) into the next time step, we have four possible combinations of $R(t_{i+1})$ and $Y(t_{j+1})$. Since R and Y are independent, the probability attached to each branch is simply the product of the individual probabilities. Let p denote the upward jump probability for $R(t_i)$ and q the upward jump probability for $Y(t_j)$. Then the four branches can be characterized as follows:

Diagram 1. Branching Process



	Event		Probability
node 1	$R(t_i)$ down,	$Y(t_j)$ down	$p_1 = (1 - p)(1 - q)$
node 2	$R(t_i)$ up,	$Y(t_j)$ down	$p_2 = p(1 - q)$
node 3	$R(t_i)$ up,	$Y(t_j)$ up	$p_3 = pq$
node 4	$R(t_i)$ down,	$Y(t_j)$ up	$p_4 = (1 - p)q$

The procedures for calculating the jump probability for a single binomial variable are well discussed in Nelson and Ramaswamy (1990). For brevity, we will not repeat them here. Instead, we briefly outline the calculation of p and q in Appendix 1.

Once the three-dimensional lattice is in place, we can work backwards along the lattice and solve for the American option's value. Since the lattice is built based on R and Y , we need to convert them back into r and S at each node according to the following formulas:

$$r = \frac{R}{\sigma_r} \quad (7)$$

$$S = \exp\left(\sigma_r \sqrt{1 - \rho^2} Y + \rho \sigma_r R\right)$$

For instance, at node (i, j) , we first do the conversion according to equation 7 to get $r(i, j)$ and $S(i, j)$. We then use $r(i, j)$ to discount back the probability weighted option values at nodes 1, 2, 3, and 4. Finally, we compare this discounted option value with the intrinsic value at node (i, j) and take the bigger of the two as the option's value at node (i, j) . Symbolically, these calculations can be summarized in the following deduction equations:

$$f(i, j) = \max(S(i, j) - K, [p_1 f_1(i+1, j+1) + p_2 f_2(i+1, j+1) + p_3 f_3(i+1, j+1) + p_4 f_4(i+1, j+1)] e^{-r(i, j)\Delta t})$$

for a call option; and

$$f(i, j) = \max(K - S(i, j), [p_1 f_1(i+1, j+1) + p_2 f_2(i+1, j+1) + p_3 f_3(i+1, j+1) + p_4 f_4(i+1, j+1)] e^{-r(i, j)\Delta t})$$

for a put option.

In the above, $f(i, j)$ denotes the option's price at node (i, j) and K is the exercise price. This backward deduction process is repeated for all nodes on the lattice until the very first node [i.e., node $(1, 1)$] is reached. The final option value at this node is then the American option's value.

This completes the illustration of the example. To show the accuracy of this numerical procedure, we will use the above stock price and interest rate processes to calculate European put option prices, which are compared against the true values based on Merton (1973).³ According to Merton (1973), a European put option's price with a stochastic interest rate can be written as:

$$P = KB(t, T)N(-d_2) - SN(-d_1) \quad (8)$$

where K is the exercise price, $B(t, T)$ is lognormally distributed and models the price of a discount bond which pays one dollar at the option's maturity date T , $N(*)$ is the probability function for a standard normal variable, and:

$$d_1 = \frac{\ln \frac{S}{KB(t, T)} + \frac{\sigma^2}{2}(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = d_1 - \sigma\sqrt{T - t}$$

$$\sigma^2 = \frac{1}{T - t} \int_t^T [\sigma_s^2(\tau) - 2\sigma_{sB}(\tau) + \sigma_B^2(\tau)] d\tau$$

In the above, σ_B denotes the volatility of the discount bond price and σ_{sB} denotes the covariance between the stock price and the discount bond price. Since the interest rate we assumed is normally distributed, the discount bond price based on this interest rate will be lognormally distributed. Therefore we could directly calculate the option's price using the interest rate process parameters. Appendix

Table 1
Comparison of Formula Prices and Numerical Procedure
Prices for European Put Options

$r = 0.1, \sigma_s = 0.25, \sigma_r = 0.08, \rho = -0.2, \mu = 0.1$
 $k = 0.5, \lambda = -0.2, K = \100

T - t	S / K = 0.9			S / K = 1.0			S / K = 1.1		
	True	Tree	American	True	Tree	American	True	Tree	American
0.2	\$9.372	9.371	10.205	\$3.484	3.486	3.687	\$0.900	0.901	0.937
0.4	9.426	9.427	10.711	4.408	4.410	4.849	1.749	1.750	1.884
0.6	9.465	9.466	11.161	4.953	4.956	5.637	2.346	2.349	2.606
0.8	9.476	9.477	11.552	5.324	5.326	6.248	2.795	2.798	3.191
1.0	9.470	9.470	11.896	5.596	5.598	6.752	3.148	3.151	3.685
1.2	9.455	9.454	12.206	5.808	5.810	7.186	3.438	3.440	4.118
1.4	9.435	9.435	12.488	5.979	5.980	7.570	3.682	3.684	4.503
1.6	9.415	9.414	12.749	6.122	6.123	7.915	3.894	3.896	4.854
1.8	9.395	9.394	12.992	6.245	6.246	8.231	4.080	4.081	5.175
2.0	9.377	9.376	13.220	6.353	6.353	8.522	4.245	4.248	5.473
2.2	9.360	9.358	13.434	6.448	6.449	8.794	4.396	4.396	5.752
2.4	9.345	9.343	13.637	6.535	6.535	9.048	4.533	4.534	6.015
2.6	9.331	9.328	13.830	6.613	6.613	9.287	4.658	4.659	6.262
2.8	9.317	9.315	14.014	6.685	6.684	9.513	4.774	4.774	6.497
3.0	9.305	9.303	14.189	6.750	6.749	9.727	4.881	4.881	6.720

Note:

This table contains three put option prices. The column headed by "True" contains exact European put option prices calculated from Merton's formula (equation 8). The column headed by "Tree" contains European put option prices calculated by using the three-dimensional tree proposed in the text. The column headed by "American" contains the prices for American-style put options calculated from the same tree.

For each time-to-maturity, we calculate prices for three types of options: in-the-money option ($S / K = 0.9$), at-the-money option ($S / K = 1.0$), and out-of-the-money option ($S / K = 1.1$). The longest time-to-maturity is 3 years. A 150-step tree is used for all the calculations.

2 presents the exact formulas for $B(t, T)$ and σ based on equations 1 and 2. The values for the process parameters are given in Table 1. It can be seen that the proposed three-dimensional lattice gives very accurate option values in all cases. The pricing errors are all within two cents. Note also that the early exercise premium is sizable for options with a longer time-to-maturity.

III. CONCLUSION

This note outlines a simple, efficient numerical procedure that allows us to calculate American-style equity option prices with a stochastic interest rate. The procedure is based on Nelson and Ramaswamy's binomial approximation for diffusion processes. The key to the simplification is to transform the stochastic stock price process and the interest rate process so that we only have to work with two independent processes. The ease of programming and the efficiency of the algorithm result from the fact that each node on the lattice will emanate into only four branches, despite the presence of two stochastic processes.

Given the recent listings of long-term equity options such as LEAPS and the fact that long-term options are more sensitive to interest rate changes, it is imperative to have an easy numerical procedure to price equity options with a stochastic interest rate. It should be pointed out that the procedure proposed in this note is perfectly general. We could assume almost any processes for the stock price and the interest rate, as long as proper cares are exercised when transforming the processes.

APPENDIX 1 CALCULATION OF BRANCHING PROBABILITIES

When using a binomial lattice to approximate a diffusion process, the calculation of the branching probabilities is crucial. For the binomial tree to converge to the original diffusion in limit, we must ensure that, at each node, the discrete movements match the true, continuous movements. In other words, from the current node to the next, we must make sure that the first two moments are matched between the discrete and the continuous processes. Most of the time, this can be achieved by choosing the right branching probabilities for the one step up and one step down movements. However, there are situations where a single jump (up or down) cannot guarantee a well-defined probability (e.g., a single jump may lead to a negative probability), which implies that the moments cannot be matched precisely. To overcome this problem, Nelson and Ramaswamy (1990) show that we could "force" multiple jumps. With the foregoing discussions in mind, we now briefly outline the calculation of the jumps and probabilities for the processes in equations 4 and 6.

Let Δt be the time interval between two nodes. Since R and Y are unit-variance processes, the size of a single jump is $\sqrt{\Delta t}$ for both variables. Next, at each node, we need to calculate the number of jumps (up and down) required for each variable's discrete and continuous moments to be matched. Referring to

(43) and (44), or (61) and (62), in Nelson and Ramaswamy, at time t , the up and down jumps for R are calculated as follows:

$$\text{Jump}_R^{\text{up}} \equiv \begin{cases} \text{the smallest, odd, positive, integer } j \text{ such that} \\ [R(t) + j\sqrt{\Delta t}] \sigma_r - r(t) \geq k[\mu - r(t)]\Delta t + \lambda \sigma_r \Delta t \end{cases}$$

$$\text{Jump}_R^{\text{down}} \equiv \begin{cases} \text{the smallest, odd, positive, integer } j \text{ such that} \\ [R(t) - j\sqrt{\Delta t}] \sigma_r - r(t) \leq k[\mu - r(t)]\Delta t + \lambda \sigma_r \Delta t \end{cases}$$

Similarly, the up and down jumps for Y are calculated according to the following:

$$\text{Jump}_Y^{\text{up}} \equiv \begin{cases} \text{the smallest, odd, positive, integer } j \text{ such that} \\ [Y(t) + j\sqrt{\Delta t}] - Y(t) = j\sqrt{\Delta t} \geq (Y\text{drift})\Delta t \end{cases}$$

$$\text{Jump}_Y^{\text{down}} \equiv \begin{cases} \text{the smallest, odd, positive, integer } j \text{ such that} \\ [Y(t) - j\sqrt{\Delta t}] - Y(t) = -j\sqrt{\Delta t} \leq (Y\text{drift})\Delta t \end{cases}$$

where $Y\text{drift}$ is defined according to equation 6:

$$Y\text{drift} \equiv \frac{\left(\frac{\sigma_r}{\sigma_s} + \rho k\right)R(t) - \frac{\rho}{\sigma_r}(k\mu + \lambda \sigma_r) - \frac{\sigma_s}{2}}{\sqrt{1 - \rho^2}}$$

Once the jump sizes for $R(t)$ and $Y(t)$ are obtained at time t on the tree, the jump probability can be calculated as [see (32), (56) and (64) in Nelson and Ramaswamy]:

$$p = \frac{k[\mu - R(t)\sigma_r]\Delta t + \lambda \sigma_r \Delta t + R(t)\sigma_r - [R(t) - \text{jump}_R^{\text{down}}\sqrt{\Delta t}]\sigma_r}{\text{jump}_R^{\text{up}}\sqrt{\Delta t} \sigma_r - \text{jump}_R^{\text{down}}\sqrt{\Delta t} \sigma_r}$$

$$q = \frac{(Y_{\text{drift}})\Delta t + \text{jump}_Y^{\text{down}}\sqrt{\Delta t}}{\text{jump}_Y^{\text{up}}\sqrt{\Delta t} - \text{jump}_Y^{\text{down}}\sqrt{\Delta t}}$$

Finally, it should be pointed out that the jump probabilities p and q are re-calculated on each node when we move backwards along the tree. They are not constants.

APPENDIX 2 FORMULAS FOR $B(t, T)$ AND σ

Given an Ornstein-Uhlenbeck interest rate process, the discount bond price can be calculated as [see, for example, Vasicek (1977)]:

$$B(t, T) = D e^{-rE}$$

where:

$$E = \frac{1 - e^{-k(T-t)}}{k}$$

$$D = \exp \left\{ \left(\mu + \frac{\lambda \sigma_r}{k} - \frac{\sigma_r^2}{2k^2} \right) [E - (T-t)] - \frac{\sigma_r^2 E^2}{4k} \right\}$$

Realizing that the discount bond volatility is $E\sigma_r$ (by Ito's lemma) and that the correlation coefficient between S and $B(t, T)$ is $-\rho$, the volatility in equation 8 can be calculated as:

$$\sigma^2 = \sigma_s^2 + \frac{\sigma_r^2}{k^2} + \frac{2\sigma_s \sigma_r \rho}{k} - \left(\frac{\sigma_r^2}{(T-t)k^2} + \frac{2\sigma_s \sigma_r \rho}{(T-t)k} \right) E - \frac{\sigma_r^2 E^2}{2(T-t)k}$$

NOTES

1. See Cox, Ingersoll, and Ross (1985b) and Hull and White (1988) for details.
2. In general, given two unit-variance diffusion processes with instantaneous correlation coefficient ρ :

$$dx_1 = (\bullet)dt + dz_1(t)$$

$$dx_2 = (\bullet)dt + dz_2(t)$$

the process defined by:

$$x_3 = \frac{x_2 - \rho x_1}{\sqrt{1 - \rho^2}}$$

will be orthogonal to the process of x_1 . See Hull (1993, p. 429) and Hull and White (1990) for details.

3. As shown in Merton (1973), American-style equity call options should not be exercised early even if the interest rate is stochastic. Therefore, an American equity call should be priced the same as its European counterpart, which has a closed-form formula.

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